EXPO
engineering ▲ design

SCADA INTELLIGENT ELECTRONIC DEVICE (IED) SIMULATOR

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Project Background

Supervisory Control and Data Acquisition (SCADA) provides real-time monitoring, fault detection, and control of power-system devices from a central operations center. It allows utilities to oversee and operate equipment such as relays, meters, and breaker controllers, helping maintain power security and safe transmission across the grid.

The Problem

Testing SCADA systems before commissioning is essential, but comes with several challenges:

High Costs:

- Hardware, software, and licensing fees can be cost prohibitive

Limited Scalability:

- Traditional test sets are difficult to reconfigure or expand without major investments

Complex Setup:

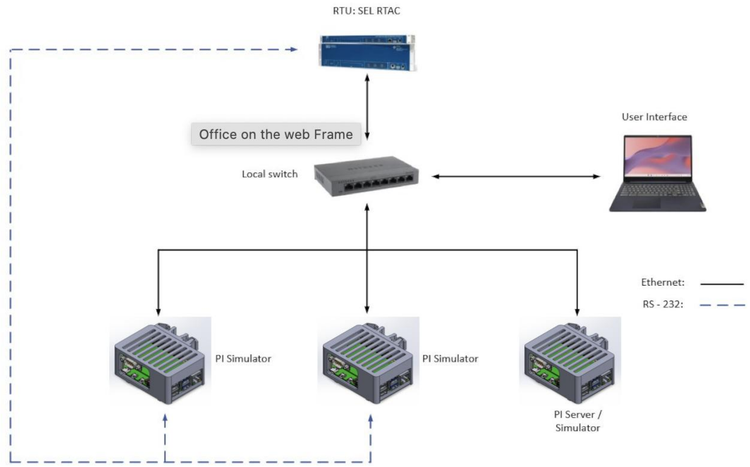
- Requires dedicated racks, wiring, power, and training

Proposed Solution

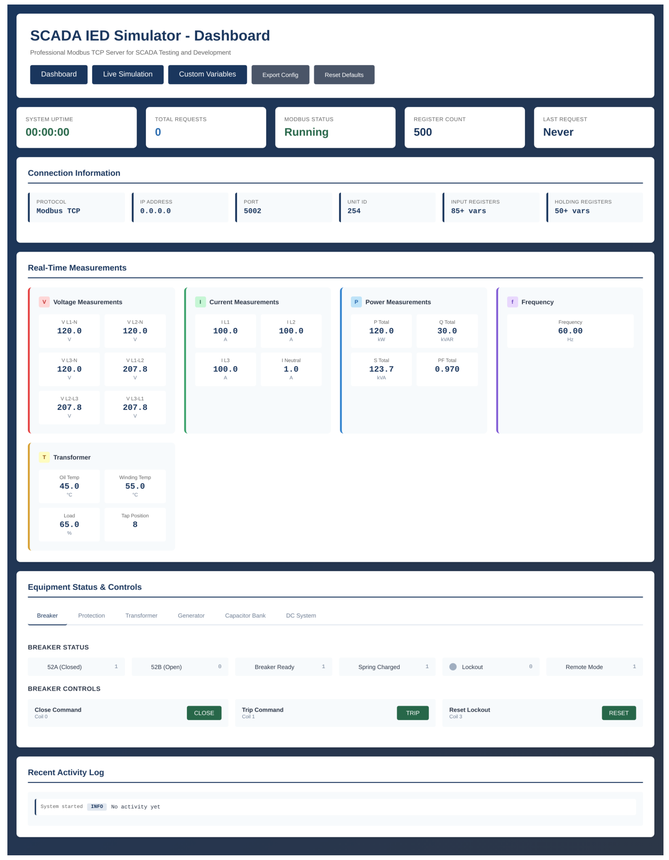
Develop an Intelligent Electronic Device (IED) simulator on a microcomputer to support SCADA testing without needing the actual hardware. The simulator will:

•Generate various data types, including analog, digital, and counters

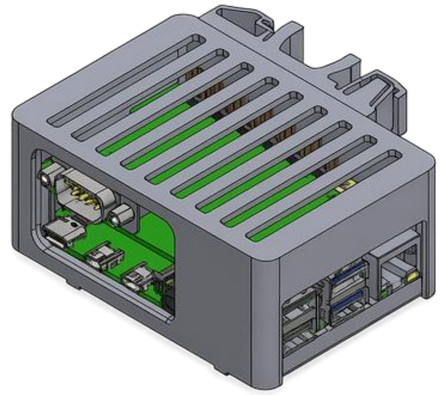
Electrical Topology



User Interface



Electronics Enclosure



The device enclosure provides easy access to all ports while supporting proper cooling and airflow. Its DIN-rail compatibility enhances configurability and ease of use in a lab setting.

Results

- Simulates analog data, digital data, and counters that can be used as replacement for an actual IED in a test environment
- A modular device that is portable and easily configurable to meet user's need for simulated data types
- Low-cost device that can be scaled to simulate multiple data at once

POWER ENGINEERS

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2025 Capstone Project